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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

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For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic Sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043-2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043-3, *Forensic Sciences – Part 3: Analysis*

ISO 21043-4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043-5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements designed to safeguard the process for the interpretation of observations including the use of statistical models and/or human judgement to address alternative propositions based on the questions asked by the customer.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A, B, C](#) and [D](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 4: Interpretation

1 Scope

This document specifies requirements and provides recommendations for the interpretation of observations in order to reach opinions that answer questions that are relevant for decision making in investigations or legal proceedings. This document states requirements that are applicable to all forensic disciplines. This document states requirements that apply when the opinion is based directly on human judgment and when the opinion is based on interpretation through a statistical model. This document is applicable to interpretation that occurs at a scene, within a forensic facility or within a judicial setting.

Interpretation is not necessary, and the requirements of this standard do not apply, if the observations resulting from the analysis directly answer the relevant question.

EXAMPLE In analytical chemistry, substances are often identified or classified. Provided that the applied analytical methods are not limited in selectivity or sensitivity for the given question, the observations can lead to a direct statement of the name of the substance (identification) or a type of material (classification). This is not considered interpretation for the purposes of this standard.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences - Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

Forensic service providers shall have documented methods for interpretation.

The interpretation methods shall be suitable for the intended use and should be supported by relevant validation studies.

Any deviation from a documented interpretation method shall be technically justified and recorded.

Interpretation shall:

- a) be supported by task-relevant data, observations, and information that are within the examiner's area of expertise;
- b) be relevant to the question put forward by the customer;

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- c) take into account all available task-relevant information;
- d) be designed to reduce the risk of cognitive bias, including designed to attempt to limit the influence of information that is not task-relevant.

Examiners shall not interpret observations where that would take them outside their area of expertise.

The examiner shall record any observations, data, information, and interpretation methods used in forming an opinion.

Known limitations and sources of error in the analysis and interpretation methods used shall be recorded and taken into account in the interpretation.

5 Questions and propositions

5.1 Questions

5.1.1 General

The examiner should not address questions concerning the intent of an accused person.

The examiner shall not address questions defined in terms of guilt or innocence.

5.1.2 Types of questions

The types of questions asked in forensic science commonly include:

- a) what is the item? (classification/identification);
- b) how much is there of an item/how many are there of an item? (quantification);
- c) who or what is the source of an item/do two or more items have the same source? (question of source);
- d) what happened, where did it happen, when did it happen, who did what? (reconstruction).

5.1.3 Classification/Identification

The classes used for classification shall be clearly defined.

If classes are defined directly by observations that do not require interpretation, the requirements of this document do not apply to that classification.

Identification of an item shall be strictly separated from individualization, which refers to question of source (see [Section 5.1.5](#)).

5.1.4 Quantification

If quantitative data do not require interpretation, the requirements in this document do not apply.

Interpretation of quantitative data shall take into account measurement uncertainty.

If a sampling method is used, the interpretation shall consider the representativeness of the sample and how the sampling method impacts the opinion given.

NOTE Depending on the situation, quantification can require interpretation. For example, interpretation would be required when estimating the total number of items in a large number of containers when individual counting is not practicable. Conversely, if all the items were individually counted, then interpretation would not be required.

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5.1.5 Question of source

When considering the source of an item, the question is either:

- a) whether the questioned-source item is from the same source as the known-source item(s) or from a different source, or
- b) whether multiple items come from the same source or from different sources.

NOTE 1 The source of the item does not imply the activity by which the item was transferred.

The interpretation of a comparison of items shall take into account the rarity (with respect to the relevant population) of the observed features of the items or the observed comparison scores.

It shall also take into account the following sources of variation with respect to observations made on questioned and known-source items:

- c) within-source variability: the variability in properties of different items from the same source;
- d) between-source variability: the variability in properties of items from different sources;
- e) measurement variability: the variability in repeated observations of the same item.

NOTE 2 It is possible for items of different sources to be similar.

5.1.6 Reconstruction

5.1.6.1 Activity

If addressing questions about activity in evaluative interpretation, for each proposition considered, the examiner shall, when appropriate, take into account the probability, if that proposition were true, that:

- a) the item(s) would be transferred from its alleged point of origin to its current location (transfer probability);
- b) the item(s) would persist in its current location given the time elapsed since it was allegedly transferred and relevant events that have occurred since that time (persistence probability);
- c) the item(s) would be recovered from its current location (recovery probability);
- d) the item(s) would be present at its current location for reasons unrelated to the alleged activity (background probability).

5.1.6.2 Event reconstruction

If conducting event reconstruction, the examiner shall consider one or more alternative propositions or alternative explanations for the observations, addressing issues such as:

- a) What activities and events occurred?
- b) Where did they occur?
- c) When did they occur?
- d) In what order did they occur (timeline)?
- e) Who carried out the activities?

5.2 Addressing the question with either investigative or evaluative interpretation

5.2.1 General

Examiners may address questions using either investigative or evaluative interpretation.

Investigative interpretation can generate likely explanations of observations, can test these explanations with new observations, and can eliminate or re-rank the explanations.

Investigative interpretation can also lead to an estimation that explains the observations.

Evaluative interpretation should be used if a relevant alternative proposition can be formulated.

EXAMPLE 1 If the question were what speed a car was moving, the examiner would use investigative interpretation to give an estimate of the speed. If the question were whether the speed was above the speed limit or not, the examiner would use evaluative interpretation.

EXAMPLE 2 If the question were about the time since death, the examiner would use investigative interpretation to give an estimate of the time since death. If the issue were whether or not the victim died within the timeframe for which the accused has an alibi, the examiner would use evaluative interpretation.

5.2.2 Investigative interpretation

In investigative interpretation, the examiner will not have propositions at the outset.

The examiner may generate possible explanations for the observations. Those explanations may be turned into propositions for a later evaluative interpretation.

The examiner may also provide estimations which explain the observations. Estimation of a value does not involve the use of propositions.

5.2.3 Evaluative interpretation

When propositions are used to address case-relevant questions, they shall be specified before the analysis is conducted. Routinely used propositions may be specified in a relevant procedure. If propositions are changed after the analysis has been conducted, such changes shall be justified and recorded.

The examiner shall consider at least two competing propositions which shall be:

- a) relevant to answering the question posed in the case;
- b) mutually exclusive;
- c) explicit and specific;
- d) considered during analysis, interpretation, and reporting.

The propositions should be provided by the customer, agreed on by the customer, or follow directly from their request.

6 Types of opinions

6.1 General

Various types of opinions can be appropriate, depending on the questions asked, the purpose of the opinion, and the information available.

The types of opinions addressed in the next subsections are:

- a) relative probabilities of the observations given the propositions (likelihood ratios);
- b) categorical opinions on the propositions;

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- c) generated explanations for the observations;
- d) estimations.

NOTE 1 Probabilities can be assigned quantitatively using statistical models, or quantitatively or qualitatively based on professional judgment. The quantitative and qualitative assignment of probabilities is addressed in [Section 7](#).

NOTE 2 The quantitative and qualitative expression of opinions is addressed in [Sections 8.2](#) and [8.3](#), respectively.

6.2 Likelihood ratios

When assigning a likelihood ratio, the examiner shall assess the ratio of the probabilities of the observations if the first proposition were true versus if the second proposition were true. If more propositions are considered, the examiner shall assess the probability of the observations under each pair of propositions and assign their ratios.

The value of a likelihood ratio may be assigned and expressed quantitatively or qualitatively.

The examiner shall not confuse the likelihood ratio with the ratio of the probabilities of the propositions.

NOTE 1 The ratio of the probabilities of the propositions, given the observations, is known as the “posterior odds”. The likelihood ratio is not the same as the posterior odds. Confusing a likelihood ratio with the posterior odds is known as the fallacy of the transposed conditional, also referred to as the “prosecutor’s fallacy”.

See [Annex B](#) for further information about likelihood ratios.

NOTE 2 A likelihood ratio can be assigned quantitatively using statistical models, or quantitatively or qualitatively based on professional judgment. The quantitative and qualitative assignment of probabilities is addressed in [Section 7](#).

6.3 Categorical opinions

Categorical opinions shall only be formed when, in the examiner’s judgement, it would be impossible to obtain the observations if any alternative proposition were true.

6.4 Explanations

When conducting an investigative interpretation, an examiner may generate explanations.

The examiner may rank the explanations by their probabilities.

When assigning probabilities of explanations, the examiner shall take into account not only the current observations but all available relevant observations and information in the case. When evaluating the probabilities of the explanations, the examiner shall take into account all explanations considered, all observations obtained from all relevant analyses, and any relevant case information.

6.5 Estimations

In investigative interpretation, an examiner may estimate a value. The examiner should assess the uncertainty of the estimation, e.g., in the form of a distribution, or a range of values.

7 Assigning probabilities

7.1 General

Probabilities are assigned using observations, existing data, and/or statistical models. Subjective assignments of probabilities may be used in the absence of this empirical information. Subjective probabilities should be clearly expressed.

The examiner shall document how probabilities are assigned, including the basis for such assignments. The examiner shall be transparent about the sources of information used to assign probabilities.

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Any assumptions used to assign probabilities should be clearly recorded.

Data used for assigning probabilities shall be as representative as possible of the relevant population of items and as representative as possible of the conditions of the case.

7.2 Statistical-model-based assignment of probabilities

Statistical models and data used to assign probabilities shall be suitable for their intended use.

Data shall only be used for assigning probabilities if, in the examiner's judgement, they are sufficiently representative of the relevant population and the conditions of the case.

The data used to assign probabilities should be from published sources or from unpublished sources that can be made available. If the data used to assign probabilities are from unpublished sources, they should be made available upon request.

7.3 Human-based assignment of probabilities

When assigning quantitative or qualitative probabilities, examiners shall use professional judgment in conjunction with relevant observations, relevant information, and relevant data.

8 Expression of the opinion

8.1 General

The opinion and the basis for the opinion shall be recorded.

The opinion may be expressed as:

- a) a quantitative or qualitative likelihood ratio;
- b) a quantitative or qualitative probability of the observations under each proposition (when more than two propositions are considered);
- c) a categorical opinion on the propositions;
- d) generated explanations of the observations;
- e) an estimation.

The opinion expressed shall explicitly refer to any propositions considered.

If no opinion can be expressed, the reasons shall be recorded.

All opinions involve professional judgement and are not statements of fact.

8.2 Quantitative expression of opinion

If a numerical value (e.g., a probability or a likelihood ratio) is calculated using quantitative observations, quantitative data, and a statistical model, the opinion should be expressed using that numerical value.

Estimations shall be expressed numerically and should be accompanied by an expression of the uncertainty of the estimate.

8.3 Qualitative expression of opinion

Qualitative opinions shall be drawn from a predefined opinion scale on which each level has a verbal expression. Examples of opinion scales are given in [Annex C](#).

The verbal expressions on the opinion scale should be defined using ranges of numerical values.

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If the interpretation method does not produce a numerical value, a verbal expression shall be recorded. The opinion scale should not be used to convert a verbal expression to a numerical value.

8.4 Categorical opinions

Categorical opinions shall not be expressed as if they were statements of fact.

Categorical opinions should be checked. This check should include examination of the items involved.

Forensic service providers shall determine when to perform a check of categorical opinions based on risk assessment. If input is received from customers, it shall be taken into account when considering risk. Records of the risk assessments shall be retained.

NOTE The risk-based decision to determine the proportion of cases requiring a check of categorical opinions based on human judgment can include factors such as the potential for an error, the impact of the types of reporting errors and experience of the examiner. The proportion of cases that require a check of categorical opinions based on human judgment can be different within and between disciplines.

The checks shall be performed by a second competent examiner in the same area of expertise and shall be recorded.

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Annex A (informative)

The forensic process

In [Figure A.1](#) below, is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

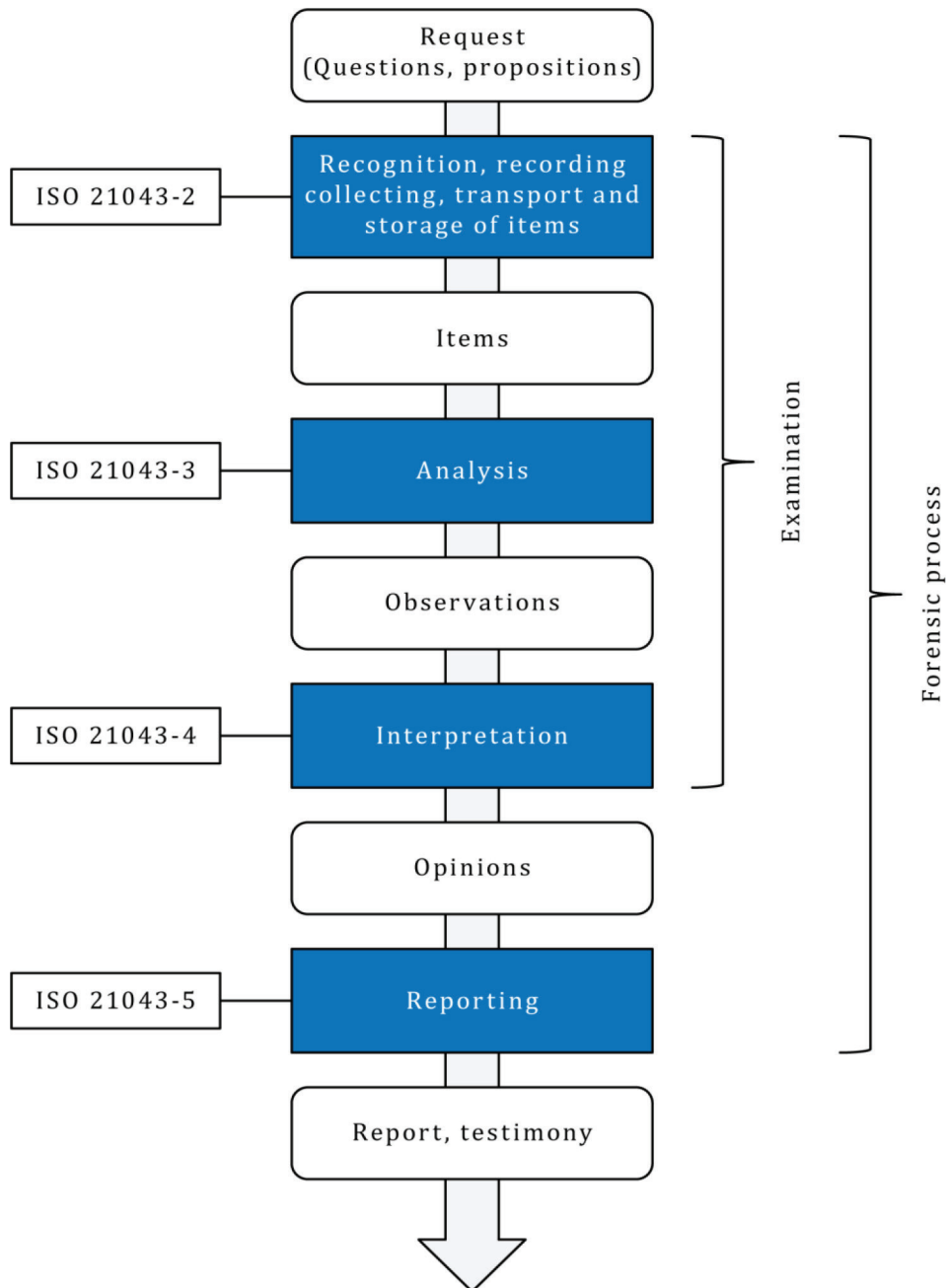


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

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The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

The meaning of a likelihood ratio

This annex is provided in order to explain the meaning of a likelihood ratio. It is not intended to instruct a court of law how to reason. In the text below, a “decision maker” is anyone who wants to make use of a likelihood ratio presented by an examiner. For the purpose of explaining the meaning of a likelihood ratio, we assume that the decision maker accepts the likelihood ratio presented by the examiner and makes use of that likelihood ratio.

An examiner uses a likelihood ratio as an expression of their assessment of the relative probabilities of the observations if one of two competing propositions were true versus if the other proposition were true. Hereinafter these will be referred to as proposition *A* and proposition *B*.

A likelihood ratio equals the probability of the observations if *A* were true divided by the probability of the observations if *B* were true. The purpose of a likelihood ratio is to express the probative value of the observations supporting proposition *A* or *B*. If a decision maker accepts the likelihood ratio presented by the examiner, that likelihood ratio expresses the amount by which the decision maker would, in light of the observations, update their belief as to the relative probability that *A* is true versus that *B* is true. This is formally expressed by Bayes’ Theorem as:

The prior odds multiplied by the likelihood ratio equals the posterior odds.

The prior odds quantify the decision maker’s belief that *A* is true divided by their belief that *B* is true *before* the examiner presents their opinion. The likelihood ratio quantifies the impact of the observations on the odds of propositions *A* and *B*. The posterior odds quantify the decision maker’s belief that *A* is true divided by their belief that *B* is true *after* taking into account the examiner’s opinion.

NOTE Bayes’ Theorem is a statement of logic. It is not a theory. It can be proven from the basic laws of probability.

The decision maker considers all the evidence presented in the case. In contrast, the likelihood ratio presented by the examiner is based only on the observations within the examiner’s area of expertise and on task-relevant information, it does not depend on other evidence.

For more comprehensive introductions to the likelihood-ratio framework in general, see [1-4]. For introductions contextualized in different branches of forensic science, see, for example: for DNA[5, 6], for voice recordings[7], and for physicochemical data[8].

Annex C (informative)

Examples of opinion scales

Introduction

The following are informative examples only. Their provision here does not imply that use of any of these particular scales is either required or recommended.

Example 1

The following is an example of an opinion scale, i.e., ordinally ranked verbal expressions for describing different degrees of strength of opinion. This example is based on the ‘ENFSI guideline for evaluative reporting in forensic science’^[9]. It provides verbal expressions that describe likelihood ratios, and the meaning of each level in the scale is defined using a range of numeric likelihood ratio values.

A likelihood ratio expresses an examiner’s assessment of the relative probabilities of the observations if one of two competing propositions were true versus if the other proposition were true. In the [Table C.1](#) below, these propositions are referred to as proposition *A* and proposition *B*.

Table C.1 — Opinion scale example 1

Range of numeric likelihood ratio values	Verbal expression of strength of opinion
1 – 2	The observations are <i>approximately equally probable</i> irrespective of whether proposition <i>A</i> were true or proposition <i>B</i> were true. The observations <i>do not support</i> one proposition over the other.
2 – 10	The observations are <i>slightly more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>weak support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
10 – 100	The observations are <i>more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>moderate support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
100 – 1,000	The observations are <i>appreciably more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>moderately strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
1,000 – 10,000	The observations are <i>much more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
10,000 – 1,000,000	The observations are <i>far more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>very strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .
1,000,000 or more	The observations are <i>exceedingly more probable</i> if proposition <i>A</i> were true than if proposition <i>B</i> were true. The observations provide <i>extremely strong support</i> for proposition <i>A</i> relative to proposition <i>B</i> .

If the observations were more probable if proposition *B* were true than if proposition *A* were true, then in the example above, the order of proposition *A* and proposition *B* in the verbal expressions would be reversed.

Example 2

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[Table C.2](#) gives another example of a scale with verbal expressions for strength of opinion, applied to a question of source.

Table C.2 — Opinion scale example 2

Verbal expression	Description
Strongest support for same source	A lot of similarity and no unexplainable difference could be observed in the comparison. The examiner would not expect to make the observations in an item from a different source.
Strong support for same source	Similarity and no unexplainable difference could be observed in the comparison. The examiner has a much higher expectation of making the observations in an item from the same source than in an item from a different source.
Limited support for same source	A little similarity and no unexplainable difference could be observed in the comparison. The examiner has a somewhat higher expectation of making the observations in an item from the same source than in an item from a different source.
No support	The observations do not provide support for one proposition over the other. The examiner's expectancy for the observations of the comparison is similar for an item from the same or from a different source. This expression is not meant to be used when no comparison could be carried out.
Limited support for different source	A little difference and no specific similarity could be observed in the comparison. The examiner has a somewhat higher expectation of making the observations in an item from a different source than in an item from the same source.
Strong support for different source	Difference and no specific similarity could be observed in the comparison. The examiner has a much higher expectation of making the observations in an item from a different source than in an item from the same source.
Strongest support for different source	A lot of difference (or any unexplainable difference) and no specific similarity could be observed in the comparison. The examiner would not expect to make the observations in an item from the same source.

Further reading

For additional examples of opinion scales, see [\[9-12\]](#).

Annex D **(informative)**

Examples of the application of the Interpretation standard

This annex aims to illustrate application of the Interpretation standard in casework.

The following are informative examples only. They do not imply that the use of a particular wording or scale is required or recommended, and other wording that adheres to the requirements can be used. Additionally, the examples do not include direction on how the interpretation should be reported.

For brevity, some underlying assumptions and the full scale from which the verbal expressions were chosen were left out.

Handwriting example (question of source, evaluative interpretation, qualitative likelihood ratio)

In a fraud case, the question was whether a questioned signature was authentic. Reference signatures were obtained for comparison by a handwriting examiner. The propositions are: (Proposition *A*) the questioned signature was written by the same person as the reference signatures, and (Proposition *B*) the questioned signature was written by some other person from the relevant population. The comparison of the signatures resulted in a number of observations, among which a few similarities, and differences which were within the natural variation of signatures. The examiner assessed the probability of the observations for both propositions, and expressed their opinion as: “my observations provide limited support for Proposition *A* relative to Proposition *B*”.

Forensic medicine example (reconstruction, evaluative interpretation, qualitative likelihood ratio)

In a possible child abuse case, Person X is accused of having caused head trauma to a child while the child was in Person X's care. Magnetic-resonance imaging (MRI) scans were taken of the child's head the day after the alleged abuse. Person X denies that the head trauma was caused by that person, and proposes that the head trauma was caused 2 weeks earlier when the child fell from their bicycle. The fact that the child fell from the bicycle two weeks earlier is not disputed. The propositions are: (Proposition *A*) The head trauma was caused the day before the MRI scans were taken, and (Proposition *B*) the head trauma was caused 2 weeks ago. A radiologist examined the MRI scans. The radiologist observed a mixed intensity lucent area in the left frontal area on the MRI scans. The radiologist reported that “observations are much more probable if the child had suffered bleeding due to the head trauma 2 weeks before the MRI scans were taken than if the child had suffered bleeding due to head trauma one day before the MRI scans were taken.”

Fingerprinting example (question of source, evaluative interpretation, categorical opinion)

In a burglary case, a fingerprint was found at the crime scene, and the question was whether it originated from Person X. The fingerprints of Person X were taken for comparison with the fingerprint. The propositions are: (Proposition *A*) the fingerprint came from Person X, and (Proposition *B*) the fingerprint came from some other person in the relevant population. The fingerprint examiner found many differences between the fingerprint and the friction ridge detail in Person X's fingerprints. In the examiner's judgement these observations would be impossible if Person X was the source of the fingerprint found at the crime scene. The examiner phrased their opinion as: “In my opinion, Person X was not the source of the fingerprint. I found many differences between the friction ridge detail in the fingerprint and the friction ridge detail in Person X's fingerprints. In my opinion, it would be impossible to observe these differences if the fingerprint originated from Person X.”

Voice comparison example (question of source, evaluative interpretation, quantitative likelihood ratio)

In a fraud case, the question was whether Person X was the person speaking on a telephone-call recording. Recordings of telephone calls in which the speaker was known to be Person X were also available. The speakers of interest in the questioned-speaker and known-speaker recordings were all assumed to be adult females speaking the same language with the same accent. The propositions were: (Proposition *A*)

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The questioned-speaker recording and the known-speaker recordings are recordings of the same speaker, and (Proposition *B*) the questioned-speaker recording and the known-speaker recordings are recordings of two different speakers from the relevant population. The relevant population was defined as adult female speakers of the same language and accent as spoken on the recordings.

The examiner used a forensic-voice-comparison system based on automatic-speaker-recognition technology. The system analyzed acoustic properties of the voices on the recording, and entered the resulting observations into statistical models that calculated likelihood ratios. The examiner calibrated and validated the system using recordings of speakers sampled from the relevant population and recorded under conditions that reflected those of the questioned-speaker and known-speaker recordings.

The examiner reported that “the probability of obtaining the acoustic properties of the questioned-speaker and known-speaker recordings is 100 times greater if they were recordings of the same speaker than if they were recordings of two different speakers from the relevant population.”

Fiber examination example (reconstruction, evaluative interpretation, qualitative likelihood ratio)

In an assault case against Person X, the prosecution alleges that: (Proposition *A*) Person X hit the alleged victim, while the defense claims that (Proposition *B*) Person X did not hit Person Y. A fiber examiner is asked to examine the clothing of Person X and the clothing of Person Y for fiber evidence. It is not disputed that, before the alleged assault, Person X and Person Y greeted and hugged each other.

In view of these agreed case circumstances and the observed absence of fibers due to the low fiber retention of the jackets that Person X and Person Y were wearing, the examiner concludes that: “The observations are approximately equally probable, irrespective of whether Proposition *A* were true or Proposition *B* were true. The observations do not support one proposition over the other.”

DNA analysis example (question of source, evaluative interpretation, quantitative likelihood ratio)

A bloodstain was found at a crime scene, and the question was if Person X was the source of the DNA in the item. The propositions are: (Proposition *A*) The DNA came from Person X, and (Proposition *B*) The DNA came from some other, unrelated person. The examiner expressed their opinion as “the observed DNA profile is at least one billion times more probable if the DNA came from Person X than if the DNA came from some other, unrelated person.”

Fire investigation example (classification, investigative interpretation)

After a building burned down, the police consider whether a criminal investigation should be started, and ask an examiner about the cause of the fire. After examining the scene of the fire, the examiner sees three possible explanations for the fire: (Explanation *A*) an electrical malfunction, (Explanation *B*) an overheated stove while cooking, and (Explanation *C*) adding a source of ignition by a person. For the purpose of helping the police decide how to proceed, the examiner ranked the explanations for the fire by their probability. The examiner expressed the investigative opinion by stating that “based on my observations and not knowing anything else about the case, in my opinion Explanation *A* is the most probable, Explanation *C* is much less probable, and the probability of Explanation *B* is very low.”

Image analysis example (estimation, investigative interpretation)

The police want to know the speed of a car that is visible in a security camera's footage. The examiner has analyzed the frame rate of the camera that was involved, and estimated the positions of the car in the frames in which it is visible. Based on the timing and the estimated distance travelled between the frames, the examiner offered the following opinion on the speed: “based on the analyzed images and the known frame rate of the camera, my estimation of the speed of the car is 72 ± 5 km/h”.

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